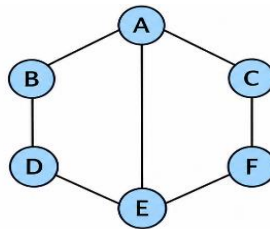


- Q1)** Write a program to implement DFS traversal for a graph. **[10 marks]**



- Q2)** Write a program to implement a Non-Linear Support Vector Machine (SVM) using the Radial Basis Function (RBF) Kernel. **[20 marks]**

Input Data :

$X = [[1, 2], [2, 3], [5, 5]]$

Class Labels :

$y = [0, 0, 1]$

Test Input : $[4, 4]$

OR

- Q2)** Write a program to implement a Basic Artificial Neural Network (ANN) for predicting outputs from input data as given below. Use different values for the hyperparameters for the learning rate and number of epochs. Find the ideal value for learning rate and # of epochs. **[20 marks]**

Learning rate: 0.1, and number of epoch = 20 (initially).

Input_vectors = $[[3, 1.5], [2, 1], [4, 1.5], [3, 4], [3.5, 0.5], [2, 0.5], [5.5, 1], [1, 1]]$

and targets = $[0, 1, 0, 1, 0, 1, 1, 0]$

- Q3)** Viva **[05 marks]**